Tutorial I (Module-I)

Subject Name: Discrete Mathematics **Subject Code: CSE-4005**

Q.1. Write the following sets in roster form:

(a) $A =$ set of all factors of 12 which are not prime (b) $A = \{x \in \mathbb{R} \mid x^2 = -1\}$

Q.2. Write the following sets in set builder form:

(a) $A = \left\{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}\right\}$ (b) $A = \left\{\frac{1}{5}, -\frac{2}{10}, \frac{3}{17}, -\frac{4}{26}, \ldots\right\}$

Q.3. Show that $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$.

Q.4. Two finite sets have m and n elements. The total number of subsets of the first set is 56 more the total number of subsets of the second set. Find the values of m and n .

Q.5. If $S, T \subseteq U$. Then show that S and T are disjoint if and only if $S \cup T = S \triangle T$.

Q.6. For any universe U and any sets $A, B \subseteq U$, the following statements are equivalent:

(a) $A \subseteq B$
(b) $A \cup B = B$
(c) $A \cap B = A$
(d) $B \subseteq A$

Q.7. Using the laws of set theory, show that

(a) $\overline{\overline{(A \cup B) \cap C} \cup \bar{B}} = B \cap C$.
(b) $(A \cap \bar{B}) \cup (\bar{A} \cap B) \cup (\bar{A} \cap \bar{B}) = (\bar{A} \cup \bar{B})$

Mention the laws used in each step.

Q.8. In a class of 25 students, 12 have taken mathematics, 8 have taken mathematics but not biology. Find the number of students, who have

(a) taken biology but not mathematics.
(b) taken biology and mathematics.

Assume that all the 25 students have at least one of the subjects.

Q.9. Draw the Venn diagram showing the following:

(a) $A \cup B \subseteq A \cup C$ but $B \subseteq C$
(b) $A \cap B \subseteq A \cap C$ but $B \not\subseteq C$

Q.10. Using Venn diagram, test the validity of the following arguments:

(a) All guilty people will be arrested. All thieves are guilty people. Therefore all thieves will be arrested.
(b) All arrested people are thieves. All thieves are guilty people. Therefore all guilty people are arrested.

GLA